



Verlet List Assignment Report

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Methodological Change

The simulation now uses a Verlet neighbor list to reduce the cost of force evaluation. Earlier, the molecular dynamics loop advanced the system and then evaluated Lennard-Jones interactions through a direct all-pairs scan. The revised formulation separates neighbor searching from force evaluation, so the interaction kernel operates only on preselected neighbors.

- The system is initialized and velocities are assigned.
- `new_verlet` builds the neighbor list after `initialize` outside the `main` loop.
- `new_verlet` is called again whenever the rebuild interval `vsteps` is reached, before the `integrate` step inside the `main` loop.
- `force_calc` uses the stored neighbor list instead of scanning all pairs.

This makes the neighbor-list update explicit rather than implicit. It also reduces the force work from global pairwise checking to a local neighborhood search.

Parameterization of the Neighbor List

The cutoff radius and the Verlet radius were taken to be equal, so that the neighbor-list search and the force cutoff use the same interaction range:

$$r_c = r_v = 10 \text{ \AA}$$

Using the three simulation sizes, the density estimate is

$$\rho = \frac{N}{L^3}$$

with the corresponding values:

Configuration	N	L (Å)	ρ (atoms Å ⁻³)	k_{est}
1600 atoms	1600	42.3212	2.11×10^{-2}	88.4
5400 atoms	5400	63.4818	2.11×10^{-2}	88.4
12800 atoms	12800	84.6420	2.11×10^{-2}	88.4

The neighbor-count estimate is obtained from

$$k_{\text{est}} = \rho \frac{4}{3} \pi r_v^3 \approx (2.11 \times 10^{-2}) \frac{4}{3} \pi (10)^3 \approx 88.4$$

The chosen capacity, $k = 200$, is therefore a conservative upper bound for the observed density.

Results From the Wall-Time Study

The timing results compare direct force evaluation with three values of the rebuild interval `vsteps`.

Method	1600 points	5400 points	12800 points
Direct pairwise evaluation	7.98	77.56	427.49
Verlet list, smaller vsteps	2.37	10.75	50.64
Verlet list, intermediate vsteps	1.79	8.34	34.88
Verlet list, larger vsteps	1.60	6.92	28.22

The Verlet-based formulation is consistently faster, and the advantage increases with system size.

Observations

The speedup is consistent with the underlying algorithmic cost. The direct formulation scales approximately as

$$O(N^2)$$

because each atom can, in principle, be compared with every other atom. In contrast, the Verlet approach reduces the force evaluation to approximately

$$O(Nk)$$

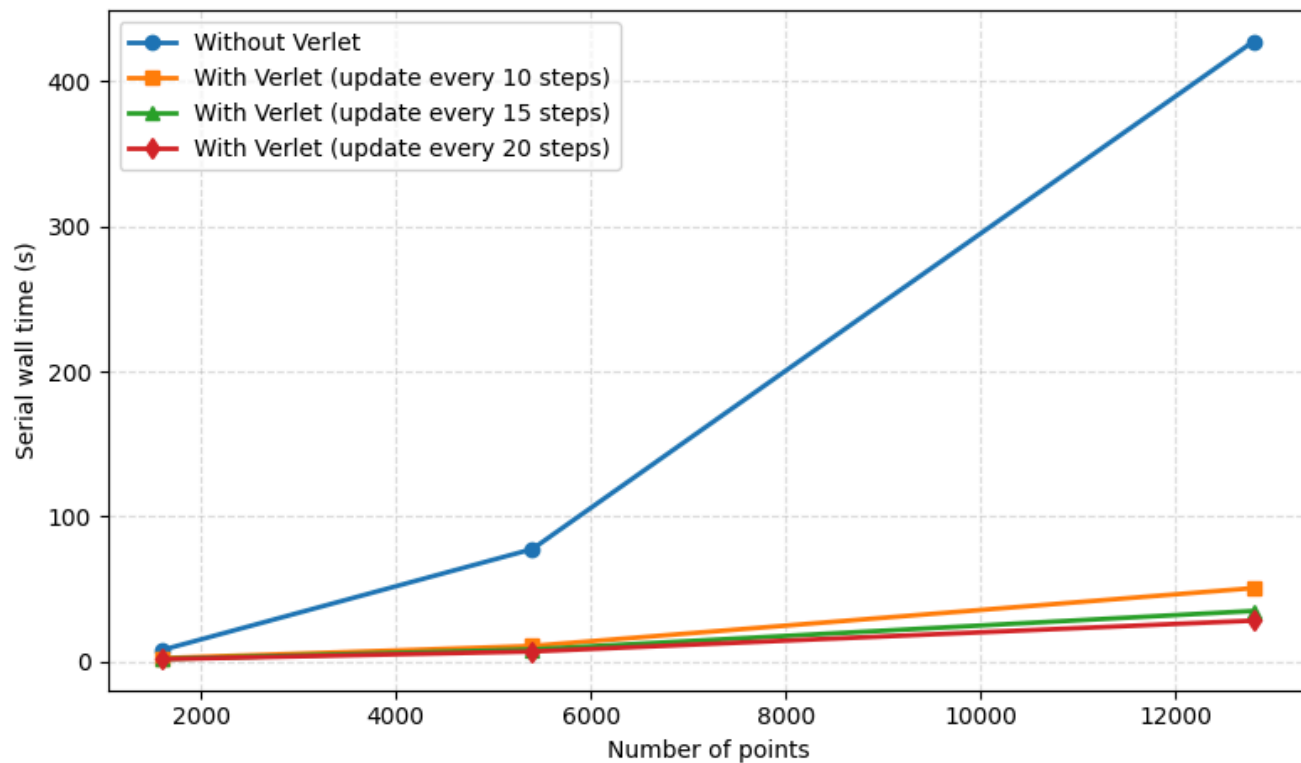
where k is the average number of stored neighbors within the cutoff region. Since k is bounded and much smaller than N , the force-evaluation cost is substantially lower. The neighbor-list rebuild cost is amortized over multiple steps and remains smaller than repeated direct all-pairs evaluation.

Among the tested cases, the larger rebuild interval produced the lowest wall time. The algorithm still uses the same physical cutoff, since $r_c = r_v$.

Figures

The first figure compares the direct and Verlet-based timings. The second figure isolates the Verlet-enabled cases, making the differences among the tested `vsteps` values easier to see.

Serial Wall Time vs Number of Points



Serial Wall Time vs Number of Points

